



INFO 442

Group 6

Data Modelling & Visualisation Report

Predicting Student Quiz Performance Using
Adaptive Interaction-Gated CatBoost

1. Introduction

Indoor environmental conditions such as CO₂ concentration, temperature, humidity and PM2.5 may influence student learning through cognitive and behavioural pathways. Building on the M4 EDA findings, this project develops predictive models for student quiz performance using environmental, cognitive, behavioural, health-related and demographic variables.

The proposed final model is an Adaptive Interaction-Gated CatBoost model (AIG-CatBoost). It combines CatBoost with interaction-based feature engineering to capture complex environmental–cognitive–behavioural relationships. Model performance is evaluated using MAE, RMSE and R², while feature importance and SHAP analysis are used to support interpretation.

2. Dataset and Feature Summary

The dataset contains 2,000 observations. The prediction target is quiz score, treated as a continuous regression outcome.

Table 1 Feature Summary	
Feature Group	Example Variables
Environmental variables	CO ₂ level, temperature, humidity, PM2.5
Cognitive variables	focus rating, cognitive impairment
Behavioural variables	reaction time, error rate
Health-related variables	asthma status, air quality label
Demographic / contextual variables	age, day, period, subject, grade

Before modelling, potential leakage variables such as performance_label and iso_outlier were removed. Categorical variables were retained for CatBoost where possible, while baseline models used encoded categorical features. The dataset was split using GroupShuffleSplit based on student ID to prevent the same student appearing in both training and testing sets.

3. Proposed Model: Adaptive Interaction-Gated CatBoost

The final model is an interaction-enhanced CatBoost regressor. CatBoost was selected because it performs well on tabular data, captures nonlinear relationships and handles categorical variables effectively.

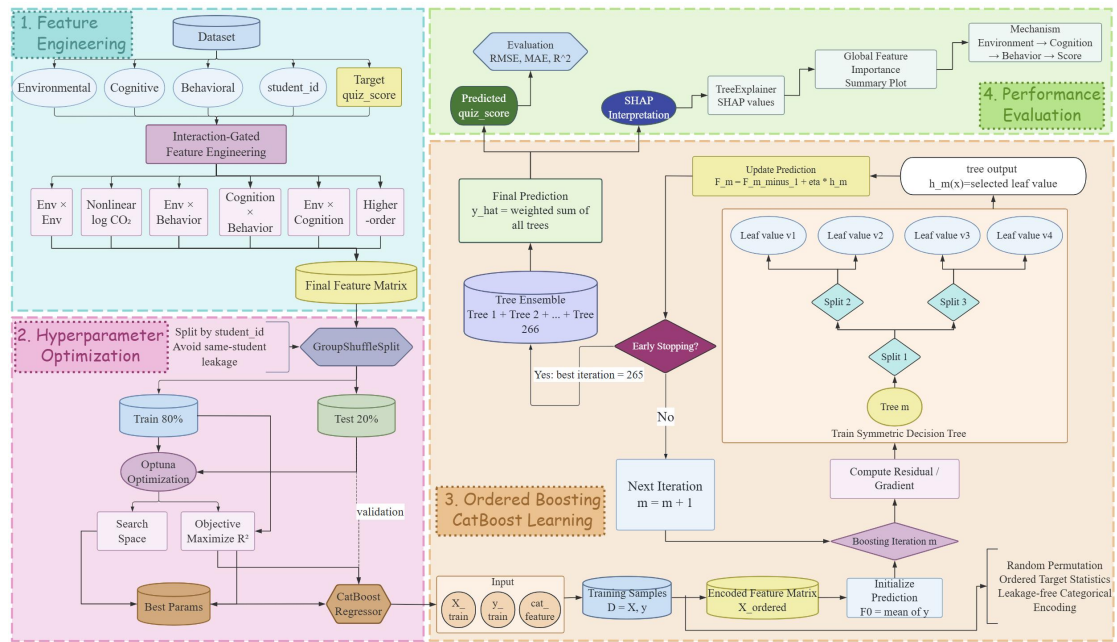


Figure 1 AIG-CatBoost Framework

The modelling pipeline includes:

1. Data cleaning and leakage-variable removal
2. Interaction and nonlinear feature engineering
3. Group-based train-test split by student ID
4. Hyperparameter tuning using Optuna
5. CatBoost training with early stopping
6. Evaluation using MAE, RMSE and R^2
7. Interpretation using feature importance and SHAP analysis

A key contribution is the construction of interaction features. Instead of modelling quiz score using isolated variables only, AIG-CatBoost captures relationships between environmental exposure, cognitive state and behavioural response.

Examples include:

Table 2 Interaction Features

Interaction Type	Example Features	Interpretation
Environmental interaction	co2_temp_new, co2_humidity_new	Combined environmental exposure

Interaction Type	Example Features	Interpretation
Nonlinear transformation	log_co2, log_pm25, co2_squared, pm25_squared	Nonlinear pollution effects
Environment-behaviour interaction	temp_rt, humidity_rt, co2_rt	Environmental impact on reaction speed
Cognitive-behaviour interaction	focus_rt, focus_error, rt_error	Attention and behavioural performance
Higher-order interaction	co2_rt_error, pm25_focus_rt	Combined environmental-cognitive-behavioural effects

4. Experimental Setup

This is a regression task because quiz score is continuous. Three metrics were used:

Table 3 Evaluation Metrics

Metric	Meaning
MAE	Average absolute prediction error
RMSE	Penalises larger prediction errors more strongly
R ²	Proportion of variance explained by the model

The final baseline experiment used 1,592 training samples and 408 testing samples after preprocessing. Optuna was used to tune CatBoost hyperparameters, including depth, learning rate, L2 regularisation, bagging temperature and random strength. Early stopping was applied to reduce overfitting.

The final AIG-CatBoost test performance was:

Table 4 Final Model Performance

Metric	Value
MAE	5.12201
RMSE	6.44481
R ²	0.76309

This means the model explains approximately 76.3% of the variation in quiz scores.

5. Baseline Model Comparison

AIG-CatBoost was compared with three nonlinear baseline models: XGBoost, LightGBM and TabNet. All models used the same train-test split for fair comparison.

Table 5 Baseline Comparison

Model	R ² Score	MAE	RMSE
AIG-CatBoost (Recommended Final Model)	0.76309	5.12201	6.44481
XGBoost Regressor	0.73815	5.46517	6.77561
LightGBM Regressor	0.73981	5.40317	6.75407
TabNet Regressor	0.73366	5.39835	6.83345

AIG-CatBoost achieved the highest R² and the lowest MAE and RMSE. This shows that the interaction-enhanced CatBoost model outperformed all baseline models.

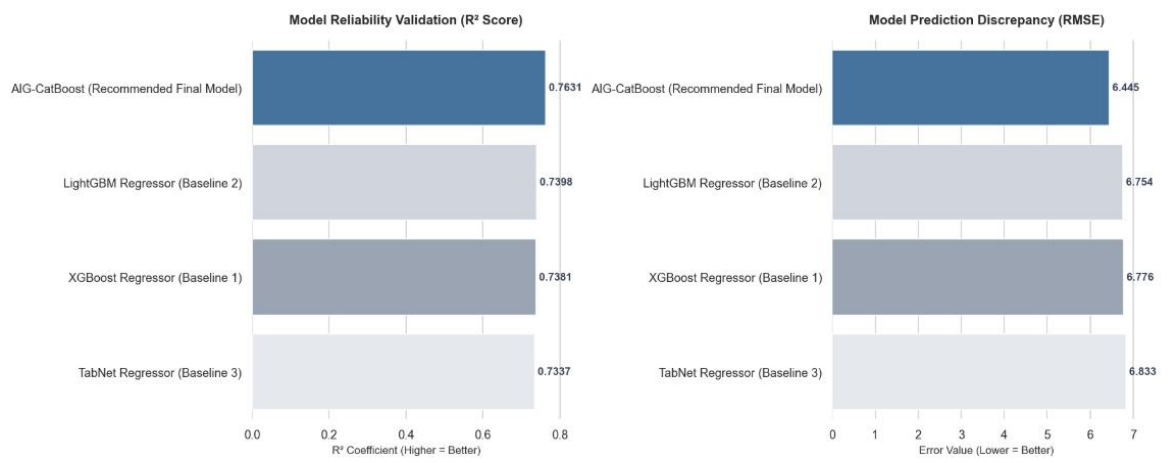


Figure 2 Performance comparison between AIG-CatBoost and baseline models using R² and RMSE

6. Ablation Study

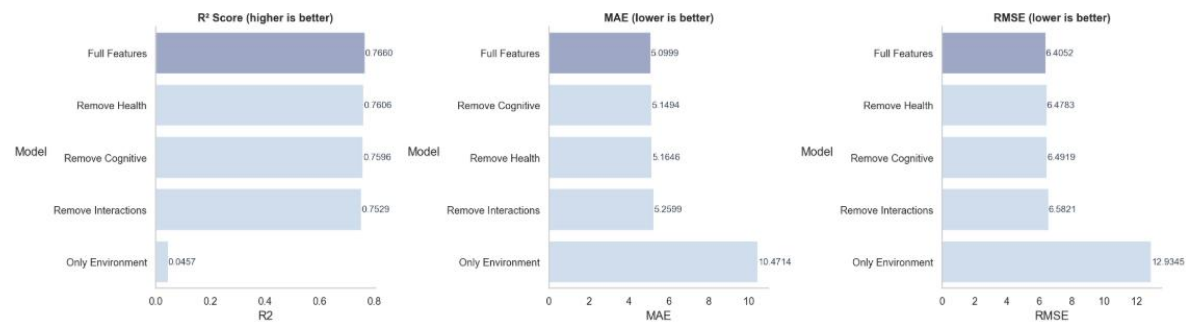
An ablation study was conducted to evaluate the contribution of different feature groups.

Table 6 Ablation Results

Model	MAE	RMSE	R ²
Full Features	5.09987	6.40522	0.76599
Remove Interactions	5.25992	6.58213	0.75289
Remove Cognitive	5.14938	6.49189	0.75962
Remove Health	5.16460	6.47831	0.76062
Only Environment	10.47137	12.93455	0.04575

The full-feature model achieved the strongest performance. Removing interaction features reduced R² from 0.76599 to 0.75289, confirming that interaction features improve prediction. The environment-only model performed very poorly, with R² = 0.04575, showing that environmental variables alone are insufficient to explain student performance.

Overall, the ablation results support the main modelling assumption: quiz performance is shaped by the combined effects of environment, cognition and behaviour.

**Figure 3 Ablation study showing the contribution of different feature groups**

7. Stakeholder Visualisations and Model Interpretation

7.1 Predicted vs Actual Scores

The predicted vs actual plot shows that most points are close to the diagonal reference line, indicating strong agreement between predicted and observed quiz scores. Larger errors mainly occur at very low and very high score ranges, suggesting that extreme cases are harder to predict.

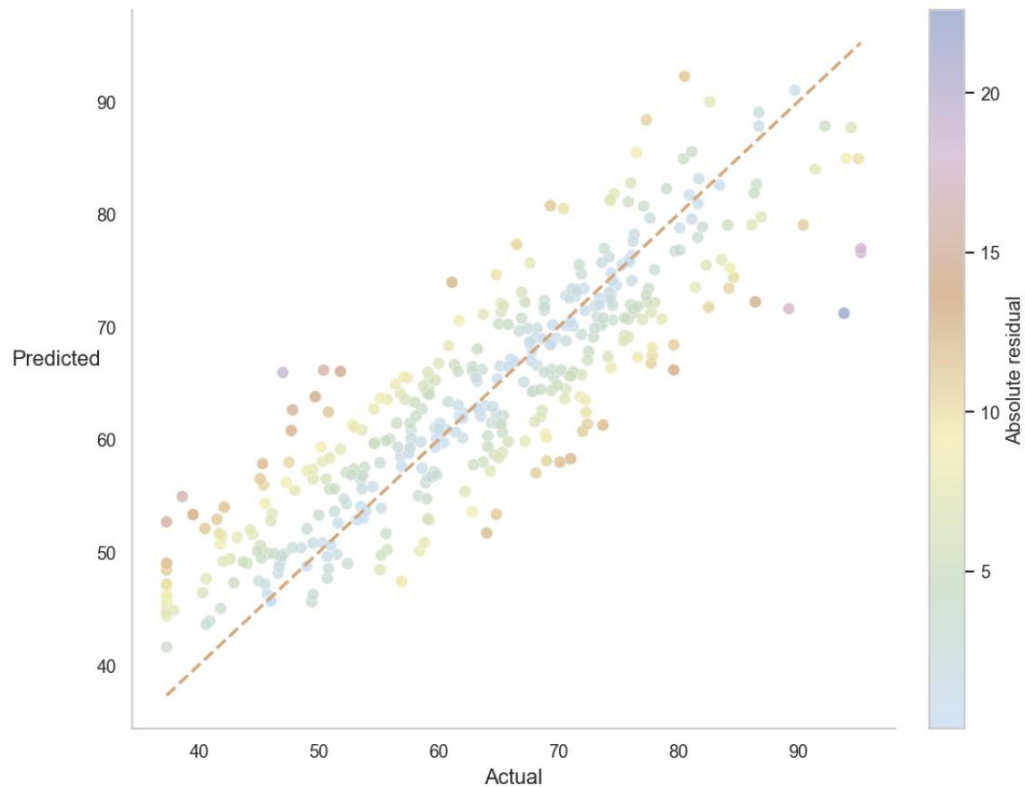


Figure 4 Predicted versus actual quiz scores for the final AIG-CatBoost model

7.2 SHAP Analysis

SHAP analysis was used to explain how features influence model predictions. The most influential features include `co2_rt_error`, `rt_error`, `humidity_error`, `error_rate` and `focus_rating`.

High reaction time combined with high error rate generally reduces predicted quiz score. Higher error rate has a strong negative impact, while higher focus rating tends to increase predicted performance. These results suggest that environmental factors may influence quiz performance indirectly through attention, reaction speed and error rate.



Figure 5 SHAP summary plot showing global feature contributions

7.3 Feature Importance

The CatBoost feature importance plot shows that many top-ranked variables are interaction features, including `co2_rt_error`, `rt_error`, `error_rate`, `humidity_error`, `focus_rating`, `co2_error` and `temp_focus`.

This confirms that interaction effects are more informative than isolated environmental measurements. For example, CO₂ alone may not fully explain quiz performance, but CO₂ combined with reaction time and error rate provides stronger predictive information.

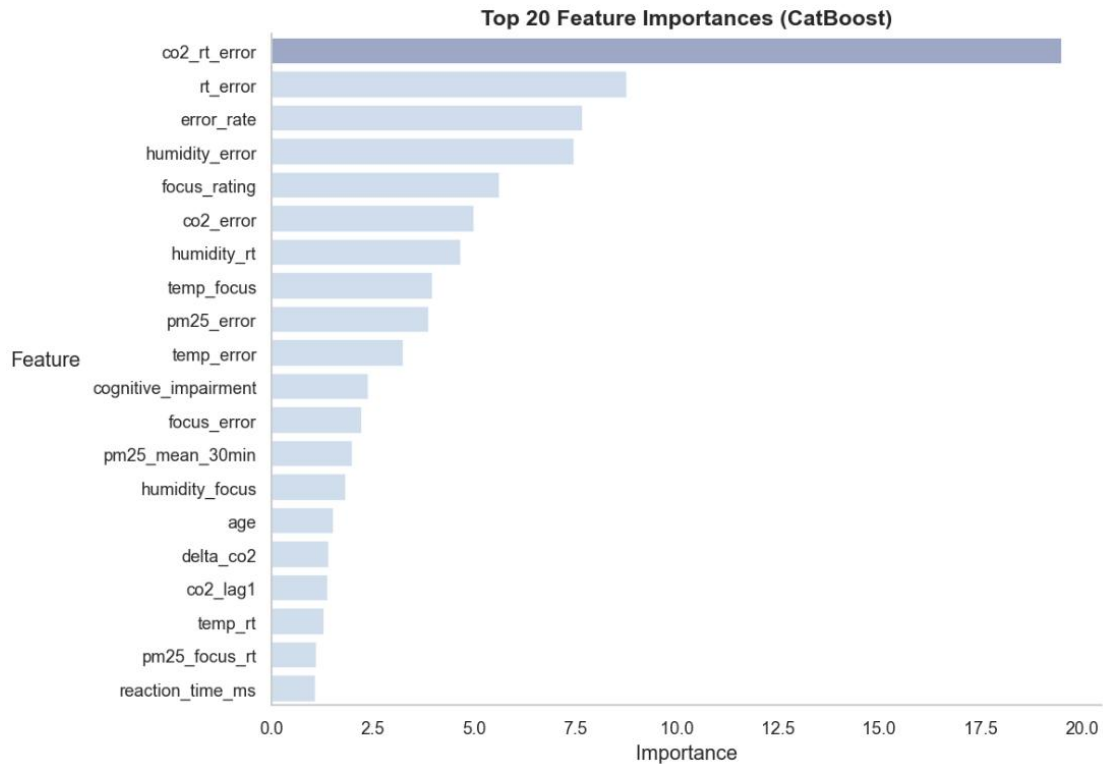


Figure 6 Top 20 feature importances derived from the CatBoost model

8. Discussion

The results show that AIG-CatBoost is the strongest model among all tested alternatives. Its superior performance is likely due to CatBoost's ability to model nonlinear tabular data and the use of interaction features derived from the EDA findings.

The ablation study further shows that environmental variables alone cannot explain student performance well. Instead, predictive power comes from combining environmental, cognitive and behavioural variables. SHAP and feature importance results support this interpretation by showing that reaction time, error rate, focus and their interactions with environmental variables are among the most important predictors.

From a stakeholder perspective, these findings suggest that learning performance should not be assessed using classroom environmental conditions alone. Behavioural indicators such as slow reaction time, high error rate and low focus may provide more actionable signals for identifying students who may need support.

9. Model Limitations and Failure Modes

Although the model achieved strong performance, several limitations remain.

First, the dataset may not fully represent all student populations or learning environments. Second, important variables such as sleep quality, prior knowledge, motivation and emotional state were not included. Third, the model performs less accurately for extreme high- and low-scoring students, as shown in the predicted vs actual plot. Fourth, the model relies on manually engineered interaction features, which may not generalise equally well to other datasets. Finally, SHAP analysis improves interpretability but does not prove causal relationships.

10. Conclusion

This project developed and evaluated an Adaptive Interaction-Gated CatBoost model for predicting student quiz performance. The model achieved the best overall performance, with $R^2 = 0.76309$, $MAE = 5.12201$ and $RMSE = 6.44481$, outperforming XGBoost, LightGBM and TabNet.

The ablation study showed that interaction features substantially improve predictive performance, while the environment-only model performed poorly. SHAP and feature importance analyses further demonstrated that reaction time, error rate, focus rating and their interactions with environmental variables are key predictors.

Overall, student performance appears to be shaped by an environmental–cognitive–behavioural mechanism rather than by environmental exposure alone. AIG-CatBoost therefore provides both accurate prediction and interpretable insights for educational data analysis.